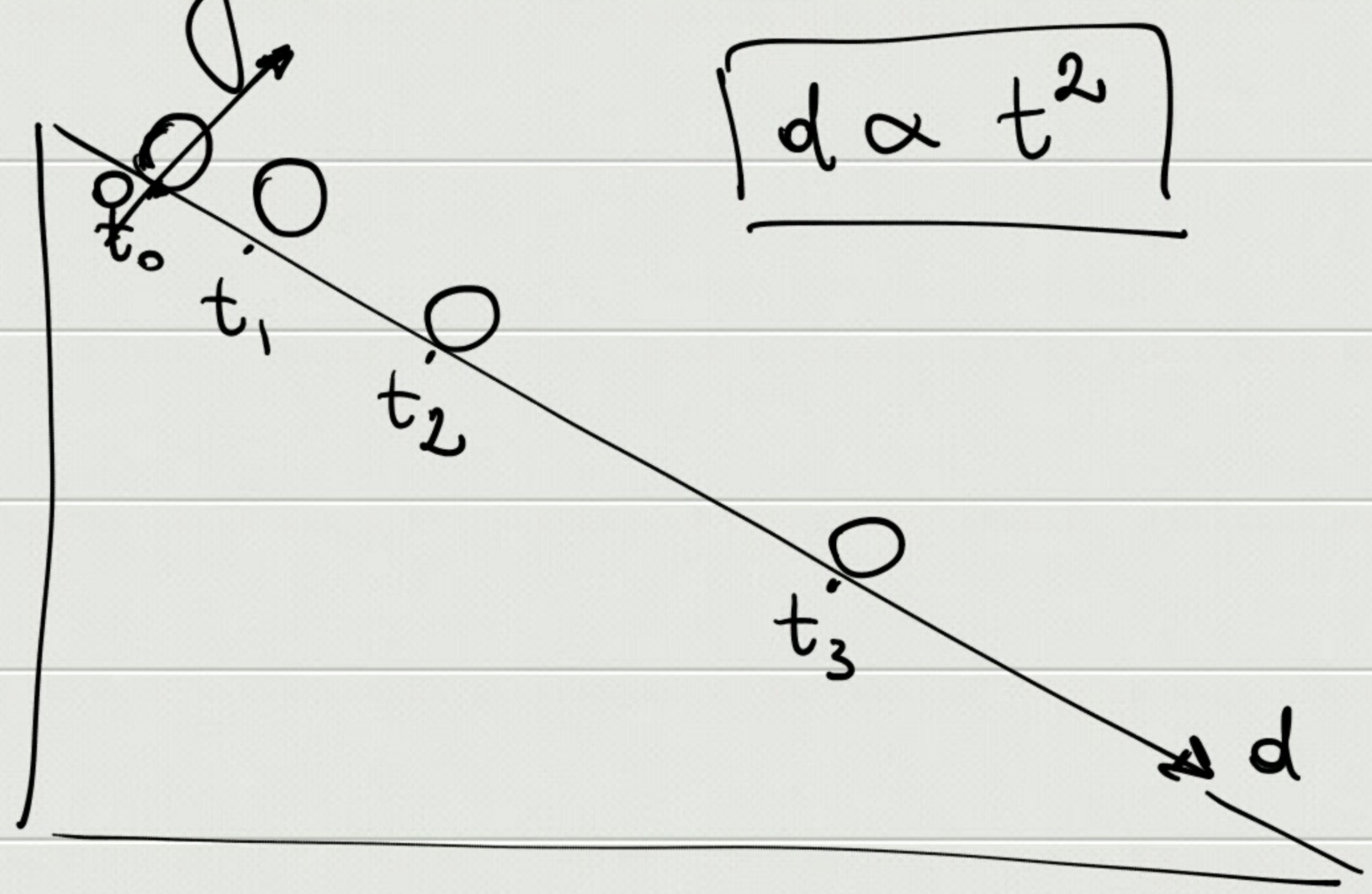
Moto di un corpo: cinematica

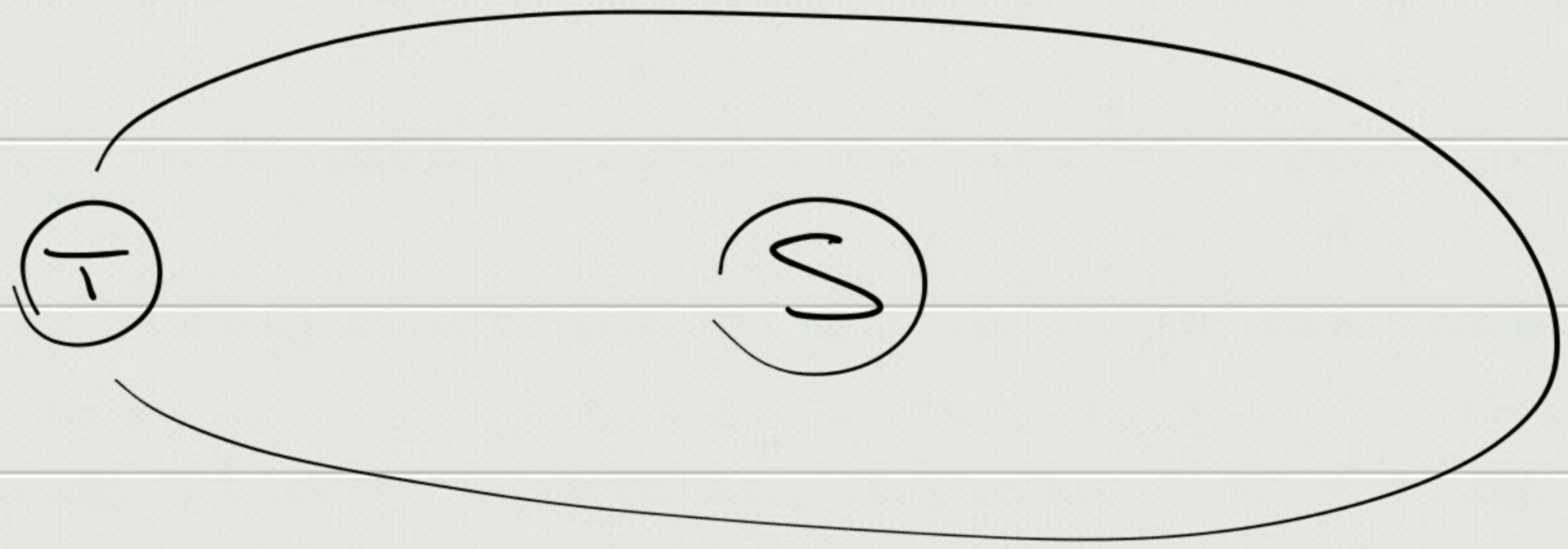
Galileo Galilei



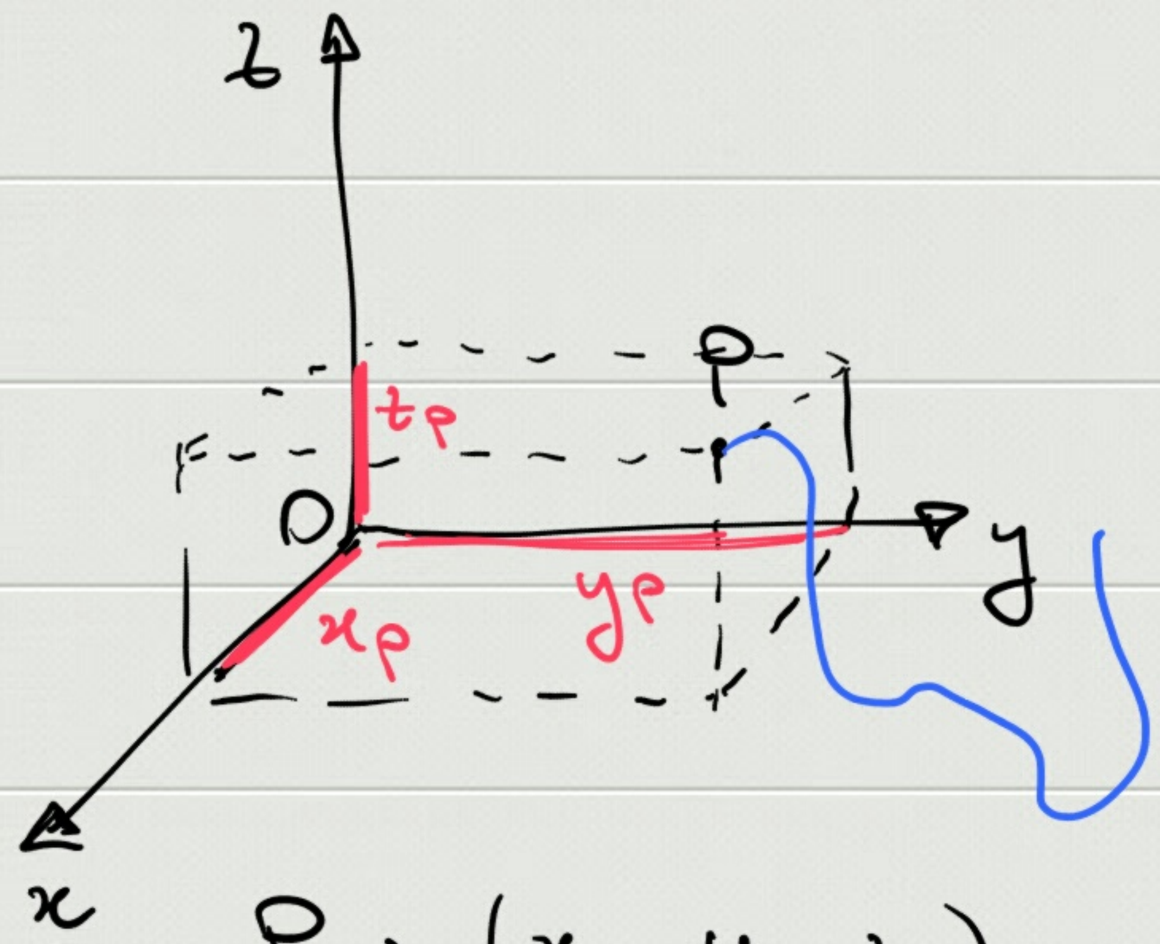
$$\boxed{d \propto t^2}$$

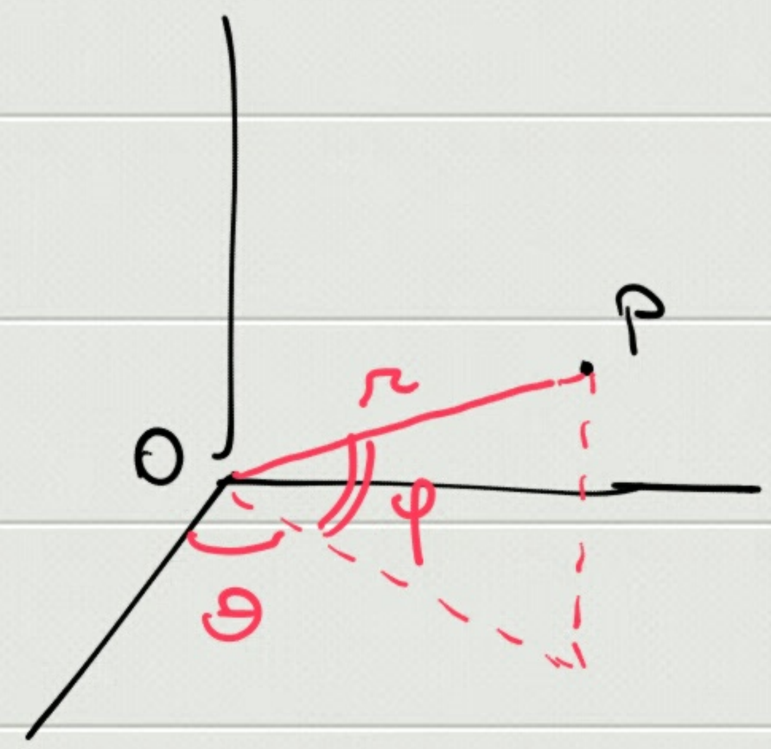
Punto materiale:

dimensioni << ambiente



Sistema di riferimento

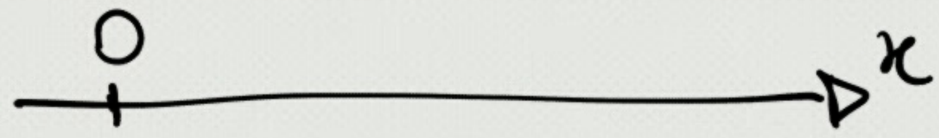




$P : (x_P, y_P, t_P)$ $\qquad$ $P : (r, \theta, \varphi)$

$P(t) : [x_P(t), y_P(t), z_P(t)]$

Moto rettilineo



t (min)	x (m)
0	0
1	400
2	1200
3	1600
4	1600
5	2500

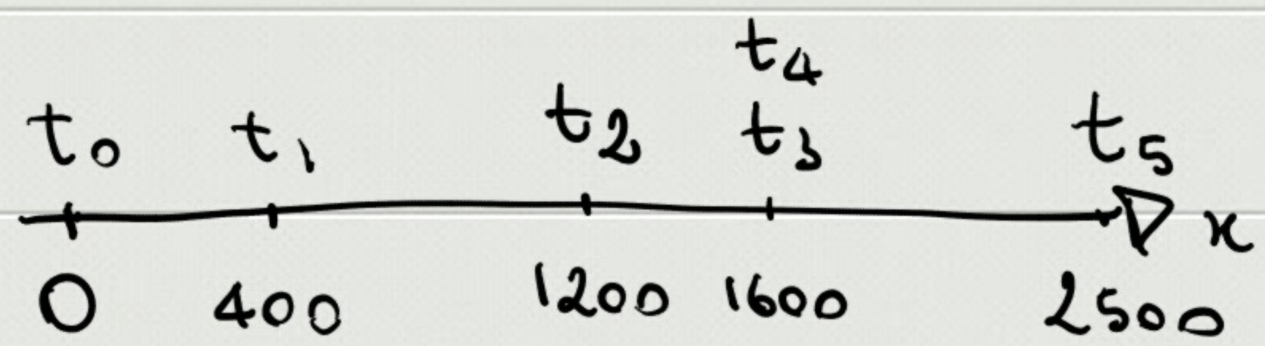
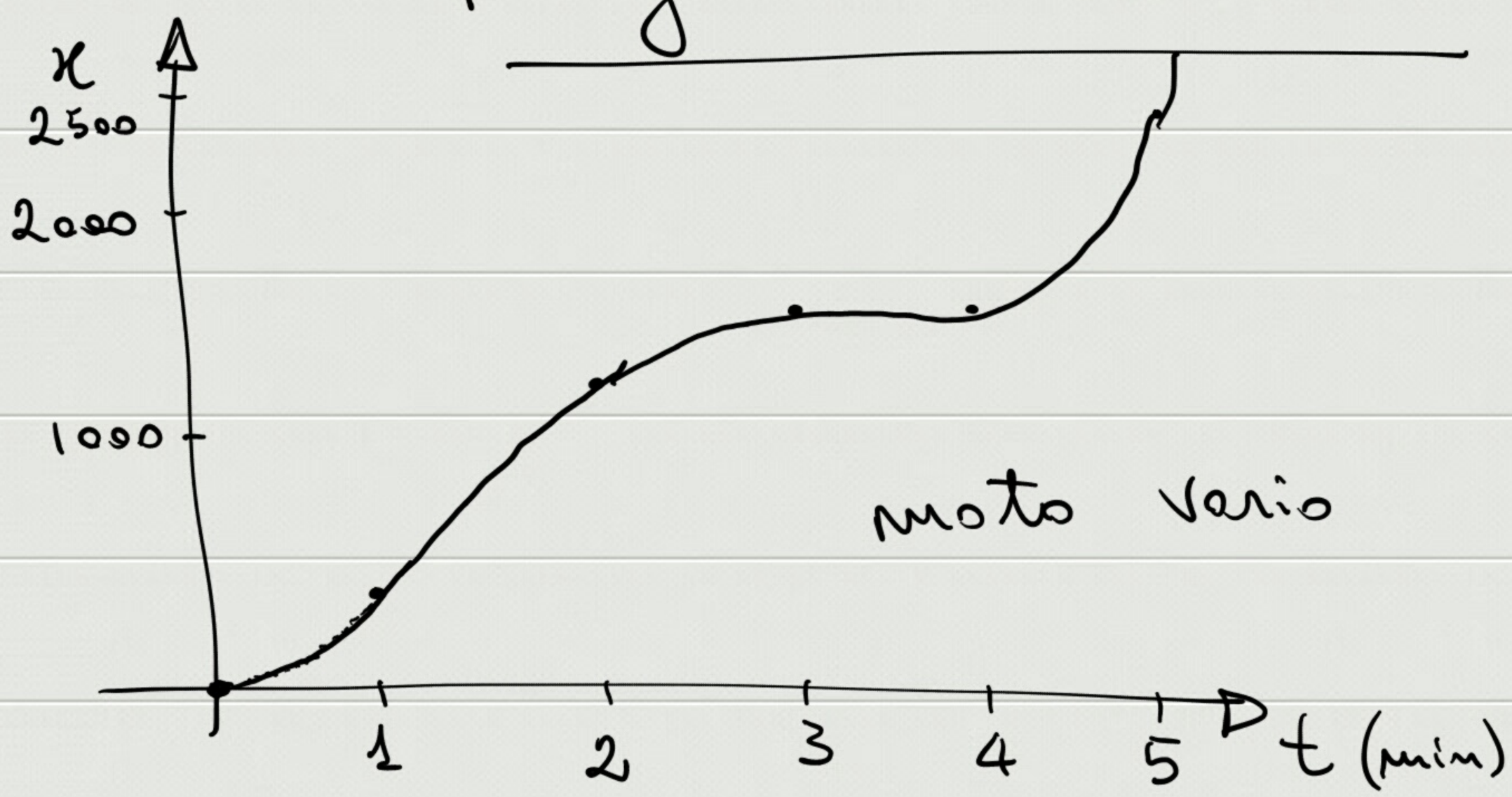
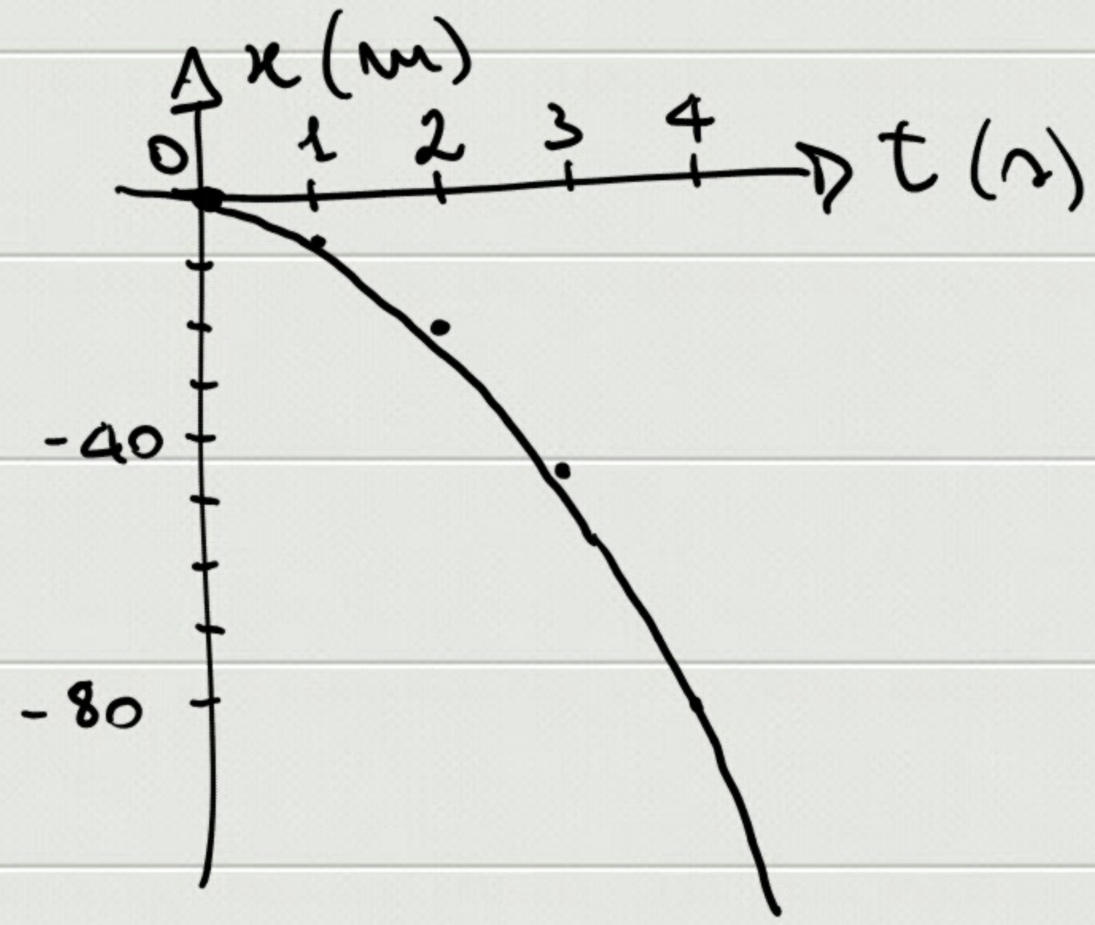


Diagramma orario



Corpo in caduta libera

t (s)	x (m)
0	0
1	-5
2	-20
3	-45
4	-80



$$\boxed{x \simeq -5t^2}$$

legge oraria del moto

$$\boxed{x = f(t)}$$